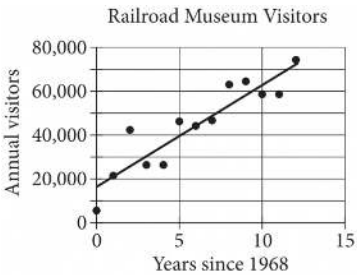


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Medium

Question



The scatterplot above shows the number of visitors to a railroad museum in Pennsylvania each year from 1968 to 1980, where t is the number of years since 1968 and n is the number of visitors. A line of best fit is also shown. Which of the following could be an equation of the line of best fit shown?

Answer

- A. $n = 16,090 + 4,680t$
- B. $n = 4,690 + 16,090t$
- C. $n = 16,090 + 9,060t$
- D. $n = 9,060 + 16,090t$

Correct Answer: A

Rationale

Choice A is correct. An equation of a line of best fit can be written in the form $y = a + bx$, where a is the y-intercept of the line and b is the slope. In the scatterplot shown, the line of best fit intersects the y-axis just over halfway between 10,000 and 20,000, or approximately 16,000. The line of best fit also intersects the graph at $(5, 40,000)$. Using the slope formula $b = \frac{y_2 - y_1}{x_2 - x_1}$ and two points that lie on the graph such as $(5, 40,000)$ and $(0, 16,000)$, the slope can be approximated as $\frac{40,000 - 16,000}{5 - 0}$, or 4,800. Only choice A has a y-intercept near the estimate of 16,000 and a slope near the estimate of 4,800. Therefore, an equation of the line of best fit could be $n = 16,090 + 4,680t$.

Choice B is incorrect because the values for the slope and the y-coordinate of the y-intercept are switched. Choice C is incorrect because the value for the slope is approximately double the actual slope. Choice D is incorrect because the values for the slope and the y-intercept are switched and because the slope is approximately double the actual slope.